HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics & Statistics

F78AB

ACTUARIAL & FINANCIAL MATHEMATICS B

Semester 2 2017/18

Duration: Two Hours

Total marks: 60

ANSWER ALL SEVEN QUESTIONS

UNIVERSITY APPROVED CALCULATOR MAY BE USED
STATISTICAL TABLES WILL BE PROVIDED

Q1 The ordinary shares of a company pay a dividend every 31 May. The next dividend is expected to be 15p per share. Dividends will grow at 4% per annum from this level and will continue in perpetuity. Inflation is assumed to be 2% per annum. The price of the share on 31 January is 570p.

Calculate the effective annual real rate of return for an investor who purchases a share on 31 January.

(Total 7 marks)

Q2 The expected annual effective return from an insurance company's investment portfolio in year t is denoted i_t. It is assumed that each i_t is independent of that in all other years and $1+i_t$ is lognormally distributed. The expected annual rate of return is 5% and the standard deviation of the annual rate of return is 9%.

(a) Calculate the expected present value of an investment of £3 million after 8 years, deriving any necessary formulae.

(6 marks)

(b) Calculate the probability that the accumulation of the investment will be more than 120% of the expected value.

(3 marks)

(Total 9 marks)

Q3 Calculate the present value of an annuity of £100 per annum payable in arrears for ten years time at the following rates of interest/discount:

(a) A rate of interest of 5% per annum convertible monthly.

(b) A rate of discount of 5% per annum convertible monthly.

(c) A force of interest of 5% per annum.

(Total 5 marks)

PLEASE TURN OVER

Q4 The force of interest $\delta(t)$ is a function of time and, for any time t, is given by the formula:

$$\delta(t) = \begin{cases} 0.02 & 0 < t \leq 5 \\ 0.004t & 5 < t \leq 10 \\ 0.003t + .0.0001t^2 & t > 10 \end{cases}$$

where *t* is measured in years.

(a) Calculate the present value of £1 due at time *t*=12.

(5 marks)

(b) Calculate the effective annual rate of interest over the 12 years.

(2 marks)

(c) Calculate the present value at time *t*=0 of a continuous payment stream that is paid at the rate of $e^{-0.05t}$ between time *t*=2 and time *t*=5.

(4 marks)

(Total 11 marks)

Q5 A small pension fund has two liabilities:

1. Annuity payments of £80,000 per annum payable annually in arrears for the next 15 years.
2. A lump sum of £100,000 to be paid in 10 years.

The trustees of the fund wish to invest in two fixed-income securities in order to immunise its liabilities. Security A has a term to redemption of 8 years and a coupon rate of 8% per annum. Security B has a term to redemption of 25 years and a coupon rate of 3% per annum. Both securities are redeemable at par and pay coupons annually in arrear. The pension fund pays no tax. The fund is currently using a rate of interest of 7% per annum effective in its calculations.

(a) Calculate the present value of the liabilities.

(2 marks)

(b) Calculate the discounted mean term of the liabilities.

(4 marks)

(c) Calculate the nominal value of each security that needs to be purchase so that the present value and discounted mean terms of the assets and liabilities are equal.

(7 marks)

(d) Without further calculation, comment on whether, if the conditions in (c) are fulfilled, the pension fund is likely to be immunised against small unform changes in the rate of interest.

(2 marks)

(Total 15 marks)

PLEASE TURN OVER

Q6 Four years ago, an investor entered into a long forward contract for a security. The contract was for 10 years. The price of the security was £90 at the inception of the contract and is £140 now. The risk-free rate can be assumed to be 3% per annum throughout the duration of the contract.

Assuming no arbitrage, calculate the value of the contract now if:

(a) The security will pay dividends of £4 in two years time and £5 in four years time.

(3 marks)

(b) The security has paid and will continue to pay annually in arrear a dividend of 2% per annum of the market price of the security at the time of payment.

(3 marks)

(Total 6 marks)

Q7 At time t=0 the n-year spot rate of interest is equal to (1.75+0.25n)% per annum effective, for $1 \le n \le 5$.

(a) Calculate the 2-year forward rate of interest from time t=3 expressed as an annual effective rate of interest.

(b) Calculate the 4-year par yield.

(c) Without performing any further calculations, explain how you would expect the gross redemption yield of a 4-year bond paying annual coupons of 3% to compare with the par yield calculated in (b).

(Total 7 marks)

END OF PAPER